Checkpoint B | Version 4 | Solutions

*Each skill earns **Pass** or **Not Yet**. Some questions contain multiple skills. Grade each skill on its own when it's part of a full question. A **Pass** is given for work showing the core skill. A **Not Yet** is given when there's not sufficient evidence of the core skill. Evidence anywhere on the submission is acceptable but evidence in the question area will be used for the initial grading. Students may request a regrade if evidence exists outside the original grading area and was not included in the original grading. Do not infer a method that the work does not support. Accept sound methods with arithmetic slips, consistent carry forward, equivalent written/graphical evidence, and minor omissions.*

Cauldrons For Days

The Standard Cauldron is a magical tool used widely in the making of various potions with many buyers and many sellers. The demand and supply curves for the Standard Cauldron can be represented by the following relationships, with prices in coins, and quantities in cauldrons.

$$D: P = 100 - \frac{1}{2}Q_d \qquad S: P = 40 + \frac{1}{2}Q_s$$

Q1 | Demand and Consumer Surplus (B1.1, B1.2)

Graph the demand curve and label the vertical intercept. Shade consumer surplus and expenditure at 80 coins.

Evidence B1.1:

Pass: Demand drawn sloping down from the right vertical intercept, with Qd read at the price and MB read at the quantity. One arithmetic slip or a missing axis label is fine when the graph and the other answer show the skill.

Not Yet: Demand sloping up, Qd and MB swapped (reading a quantity as a price), or no graph.

Evidence B1.2:

Pass: CS shaded as the triangle below demand and above the price, expenditure as the rectangle below the price, both computed. One arithmetic slip is fine when the regions are right.

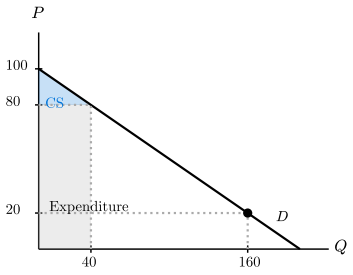
Not Yet: CS and expenditure swapped, CS measured below the price, or no areas.

a) Qd at 80: **40** (B1.1)

b) MB of the 160th: **20** (B1.1)

c) CS at 80: **400** (B1.2)

d) Expenditure at 80: **3200** (B1.2)



Checkpoint B | Version 4 | Solutions

Q2 | Supply and Producer Surplus (B2.1, B2.2)

Draw the supply curve and label its intercept. Shade producer surplus and revenue at 80 coins.

Evidence B2.1:

Pass: Supply drawn sloping up from its intercept, with Qs read at the price and MC read at the quantity. One arithmetic slip or a missing axis label is fine when the graph and the other answer show the skill.

Not Yet: Supply sloping down, Qs and MC swapped, or no graph.

Evidence B2.2:

Pass: PS shaded as the triangle above supply and below the price, revenue as the rectangle below the price, both computed. One arithmetic slip is fine when the regions are right.

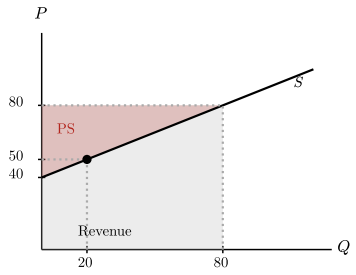
Not Yet: PS and revenue swapped, PS measured above the price, or no areas.

a) Qs at 80: **80** (B2.1)

b) MC of the 20th: **50** (B2.1)

c) PS at 80: **1600** (B2.2)

d) Revenue at 80: **6400** (B2.2)



Checkpoint B | Version 4 | Solutions

Q3 | Equilibrium and Efficiency (B3.1, B4.1)

Use a graph and algebra to find equilibrium price and quantity.

Evidence B3.1:

Pass: Equilibrium from setting quantity demanded equal to quantity supplied, the shortage or excess named and sized, and the price moving in the direction that closes it. An arithmetic slip carried through consistently is fine.

Not Yet: Shortage and excess swapped, or a price direction that would widen the gap.

Evidence B4.1:

Pass: MB and MC read at the restricted quantity, MB above MC recognized as room for more surplus, and the deadweight loss as the triangle between the curves from that quantity to equilibrium. One arithmetic slip is fine when the region is right, as is MB and MC right with the triangle set up but not finished.

Not Yet: A deadweight loss on the wrong side of equilibrium, or the whole area between the curves taken as the loss.

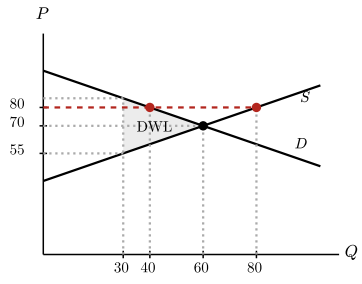
a) Equilibrium Price: **70** Equilibrium Quantity: **60** (B3.1)

b) At 80 coins: shortage, **excess** (circle one) of size **40** (B3.1)

c) From 80 coins, the price will: rise, **fall** (B3.1)

d) At 30 cauldrons, MB: **85** MC: **55** (B4.1)

e) Deadweight loss if only 30 are exchanged: **450** (B4.1)



Checkpoint B | Version 4 | Solutions

Q4 | Price Controls (B4.2)

The Ministry of Magic grew concerned that Standard Cauldrons had become too expensive for many young witches and wizards. Use algebra and a graph to evaluate whether a binding price ceiling of 60 coins would allow more students to purchase them.

Evidence B4.2:

Pass: The ceiling recognized as binding, the quantity exchanged as the smaller of Qd and Qs, the shortage sized, PS placed and computed as the triangle below the ceiling and above supply out to the quantity traded, and sellers seen to lose. One quantity slip carried through consistently is fine, as is the PS triangle right with an arithmetic slip.

Not Yet: The larger quantity used as the quantity exchanged, the ceiling treated as not binding, or PS measured above the ceiling.

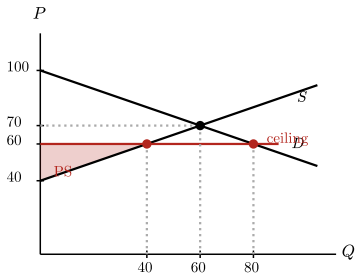
a) Quantity Demanded: **80** Supplied: **40** Exchanged: **40**

b) Excess, **Shortage** (circle one) of size **40**

c) Producer Surplus: **400**

d) Are more Standard Cauldrons being purchased? **No**

e) Compared with equilibrium, sellers: gain, **lose** (circle one)



Checkpoint B | Version 4 | Solutions

Q5 | Price Elasticity (B5.2)

Use the midpoint method with the quantities in the table, then circle the classification.

> **Evidence B5.2:**
>
> **Pass:** Both midpoint elasticities with the right classifications, demand classified by its absolute value. One arithmetic slip is fine when the method and classifications are consistent with the numbers.
>
> **Not Yet:** Slope used in place of elasticity, a classification that contradicts the number, or no midpoint averages.

a) Demand, price from 60 to 80 coins: **−7/3 elastic**, unit elastic, inelastic

b) Supply, price from 60 to 80 coins: **7/3 elastic**, unit elastic, inelastic

P	Qd	Qs
60	80	40
80	40	80

Checkpoint B | Version 4 | Solutions

Q6 | Shifters and Comparative Statics (B5.1, B5.3)

One terribly cold winter, a new mining technique increased the availability of a metal used in the production of Standard Cauldrons, making it easier to access the resources required to produce them. Afterward, the Charmfire Toaster became a popular alternative to the Standard Cauldron in many kitchens.

Evidence B5.1:

Pass: The right curve and direction for both events, with the shifted curves drawn. Up and down are read on the price axis: supply down is supply out, demand down is demand in. In, out, left, and right are accepted when they match. One direction mislabeled is fine when the graph shows the right shift.

Not Yet: A movement along a curve treated as a shift, or the wrong curve.

Evidence B5.3:

Pass: For the two shifts together, the one change that is determined and the one that is indeterminate, read from the directions of the shifts. One of the two reported the wrong way is fine when the graph shows both shifts.

Not Yet: Both changes reported as determined, or the determined change in the wrong direction.

a) Which curve shifted? **Supply** Up, **Down** (circle one) (B5.1)

b) After the toaster, which curve shifted? **Demand** Up, **Down** (circle one) (B5.1)

c) After both, Price Change: **decreased** Quantity Change: **indeterminate** (B5.3)

